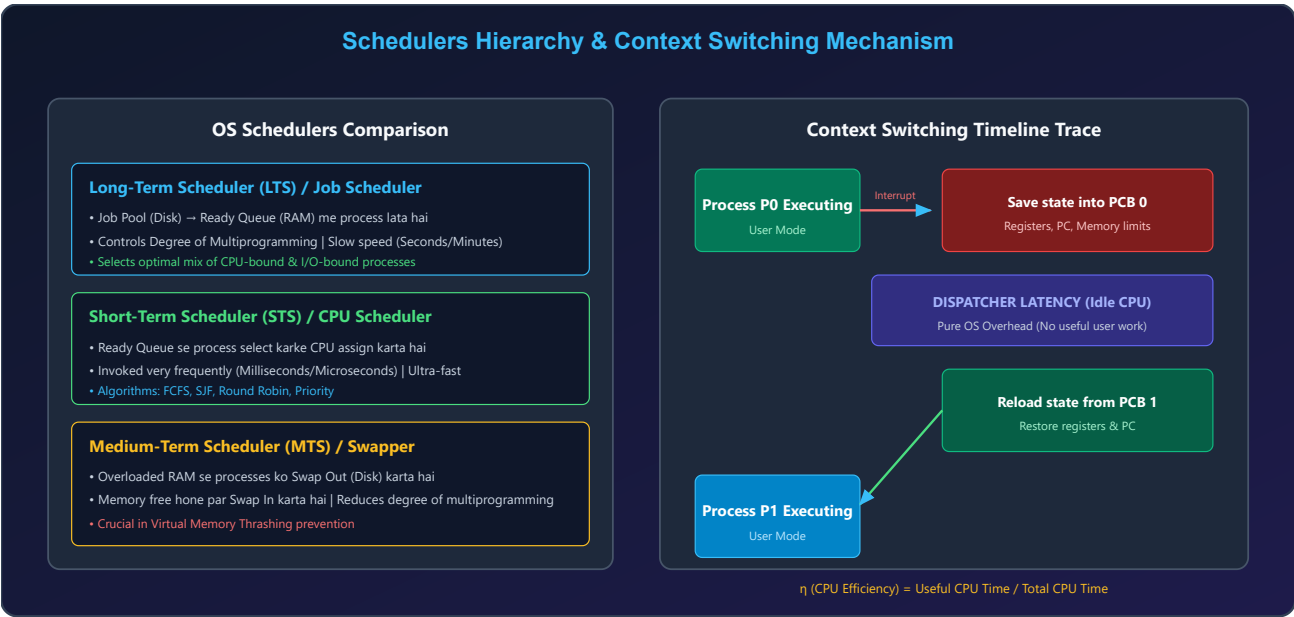


Module 02: Schedulers, Dispatcher & Context Switching

1. OS Schedulers Comparison

Feature	Long-Term Scheduler (LTS)	Short-Term Scheduler (STS)	Medium-Term Scheduler (MTS)
Other Name	Job Scheduler	CPU Scheduler	Swapper
Selection Path	Disk Pool → Ready Queue	Ready Queue → CPU	RAM ↔ Disk (Swap space)
Frequency	Very low (Minutes)	Extremely high (Milliseconds)	Moderate (Seconds)
Speed	Slowest	Fastest	Medium
Multiprogramming	Controls degree directly	No control over degree	Reduces degree of multiprogramming

2. Schedulers & Context Switch Flow



3. Context Switching & Efficiency

Dispatcher Overhead: CPU context switch ke dauran idle rehta hai.

$$\text{CPU Efficiency } (\eta) = \frac{\text{Useful Time}}{\text{Useful Time} + \text{Context Switch Overhead}}$$